

Neha Waghchoure

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[LinkedIn](#) [Github](#)

EDUCATION

- Stony Brook University** New York, US
Masters in Science - Data Science
- Indian Institute of Technology, Madras** Chennai, India
Diploma - Data Science
- Pune Institute of Computer Technology** Pune, India
Bachelor of Engineering - Electronics and Telecommunication

SKILLS SUMMARY

- Languages:** Python, C++, SQL, Java, HTML/CSS, JavaScript
- AI/ML/LLM:** LangGraph, LangChain, RAG, Fine-tuning (PEFT/QLoRA), Prompt Engineering, OCR
- Frameworks:** PyTorch, TensorFlow, FastAPI, Scikit-learn, Pandas, NumPy, Keras
- Tools/Platforms:** AWS (Textract, S3), OCI, Docker, Git, MySQL, Power BI, Excel

EXPERIENCE

- AI/ML Engineer** Mumbai, India
TIH-IIT Bombay Feb 2026 - Present
 - Leading end-to-end design and deployment of production-grade **Generative AI** pipelines.
 - Architecting scalable, low-latency backend infrastructures using **Asynchronous FastAPI** and **Docker** to host **LLM-based** solutions for on-premise inference and data privacy.
 - Optimizing data pipelines and model inference strategies on **OCI** (Oracle Cloud) and **AWS**, reducing processing costs and dimensionality by **75%**.
- AI/ML Project Engineer** Mumbai, India
TIH-IIT Bombay Feb 2025 - Feb 2026
 - Developed an independent **ETL** module on semi-structured financial data for downstream analytics.
 - Fine-tuned 3.2 B LLM** on domain specific financial documents to enhance data extraction accuracy and automate financial information retrieval.Developed logic-driven data processing pipeline for validation and computation of financial metrics.
 - Engineered and optimized large scale financial data sets for **predictive modeling**. Condensed high dimensional data from about 16K to 4K impactful features and implemented anomaly and outlier detection. Developed and deployed ML pipelines on **OCI** to support predictive settlement and collection forecasting.
- Research Intern** Pune, India
Pune Institute of Computer Technology July 2024 - Oct 2024
 - Developed a smart marketing hardware system utilizing AI-enhanced OCR and content scheduling algorithms to automate cross-platform workflows.

PROJECTS

- AI-Powered Credit Underwriting Solution:** Architected a **RAG pipeline** for financial document extraction. Integrated **AWS Textract** with LLM-based semantic retrieval to analyze complex financial data, improving data extraction accuracy from **75% to 94%**,increasing end user's productivity by **40%**, and strengthening risk frameworks for retail lending. Engineered a module to transform financial credit data into derivable insights.
- Collections-Settlements:** Optimized large-scale financial datasets for predictive settlement forecasting model. Applied feature reduction (16K to 4K features) and **unsupervised learning** for **anomaly detection**.
- Bank Telemarketing Success Prediction:** Developed a predictive pipeline using **XGBoost** and **LightGBM** on a dataset of 40k+ entries. Applied feature engineering and hyperparameter tuning to improve prediction recall for high-value customers by **15%**.
- Pet Image Classification Using CNN:** Implemented ResNet, AlexNet, and VGG architectures. Trained and evaluated models for image classification tasks using automation scripts. **Tech:** Python, PIL, TensorFlow, Keras, NumPy, Matplotlib.
- Intelligent Hybrid Steganography Optimization:** Research project on Diamond Encoding algorithm for cybersecurity. Implemented attack-proof text-in-image embedding using **TensorFlow**. Published findings in **AIP Conference Proceedings**.

CERTIFICATIONS

- **MLOps Specialization:** Duke University, Coursera (Jul 2026)
- **AWS ML Fundamentals:** Udacity Nanodegree (Sep 2025)
- **AI Programming with Python:** Udacity Nanodegree (Mar 2025)

PUBLICATIONS & PATENTS

- **Patent:** Smart Management Device for Enhanced Marketing (UK Design Right #6408730). Registered: Dec 2024
- **Survey:** Survey on Intelligent Hybrid Steganography Optimization Algorithm, AIP-ICDIDSA-23. (June 2024)
- **Publication:** Intelligent Hybrid Steganography Optimization Algorithm. (In process with Springer Publishing, 2025)

VOLUNTEER EXPERIENCE

- **Imagine-Her** Remote
Systems Dev Volunteer Oct 2024 - Feb 2025
 - Architecting digital systems to support 100+ women-led businesses in rural Uganda.
- **PICT-INC** Pune, India
Event Decor Lead April 2022 - May 2023
 - Led a team of 15 peers to manage logistics, vendor relations, and budgets for a national symposium.